

S.No.	Component Name	Qty	Description
1	Raspberry Pi Zero 2 W	1	Quad-core 64-bit Arm Cortex-A53 CPU @ 1GHz, 512MB LPDDR2 SDRAM, 2.4GHz 802.11 b/g/n wireless LAN, Bluetooth 4.2 / BLE, Mini HDMI, micro USB OTG, MicroSD slot, CSI-2 camera connector, 40-pin GPIO header footprint.
2	MicroSD Card	1	64GB San Disk SD Card
3	3.5-inch TFT Resistive Touch Screen LCD	1	480x320 resolution, SPI interface , resistive touch control (XPT2046 controller), ILI9486/ILI9488 driver IC, 5V/3.3V compatible, designed for Raspberry Pi GPIO.
4	Lithium Ion Polymer Battery	2	3.7V 2600 mAh (In Parallel)
5	18650 Battery Holder	2	Battery holder for 18650 Li-ion cells
6	1S 20A BMS	1	1S 20A Battery Management System for 18650 cells to prevent Overcharge/Discharge
7	TP4056 Battery Charging Module	1	5V 1A Type-C lithium battery charging board with protection circuit.
8	MT3608 DC-DC Step Up Boost Module	1	Input 2V-24V, Output 5V-28V
9	Voltage Sensor Module 0-25V	1	Measuring Voltage of Battery
10	SPDT Slide Switch	1	Micro slide switch for main power control (ON/OFF).
11	PCM5102 I2S DAC Module	1	I2S interface, Analog Output
12	PAM8403 Audio Amplifier Module	1	3W+3W dual channel Class-D audio amplifier, 5V power supply.
13	Mini Speaker	2	8 Ohm, 0.5 Watt, 18mm
14	10K Ohm Thumbwheel Potentiometer	1	Volume control dial
15	Tactile Push Button Switches	13	6x6x5mm, for Buttons (A,B,L,R etc.)
16	MCP23017 GPIO Expander	1	16-bit I/O expander with I2C interface
17	Common Cathode RGB LED	2	Common Cathode RGB LED for battery/charging indication
18	PCF8591 ADC Module	1	Analog to Digital Convertor
19	Outer Shell	1	Custom 3D Printed
20	Miscellaneous Components	-	Resistors, Capacitors, Transistors, Diodes, Wires, Solder, Heat Shrinks, Perf Boards and other generic electronics components.